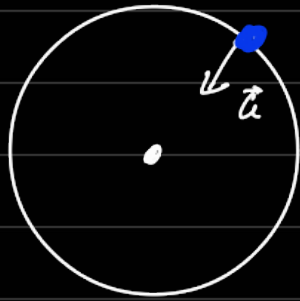
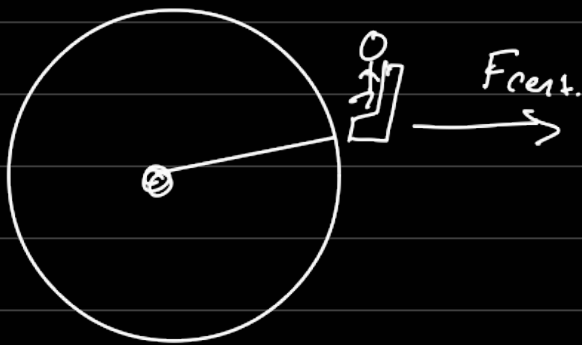


* So, we found the centripetal acceleration
 $\vec{a}$



• This appears as a fake/fictitious force when the person at $\bullet$ experiences this motion

• From the point of view of the person $\bullet$ moving around the circle, they "experience" a force called the centrifugal force: which they feel pulling them out



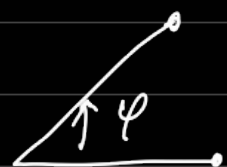
* Although we didn't get to the point of seeing how we discussed this in terms of relative experience.

→ Here I'll go over the maths to show how.
From this, we'll be able to understand/do
(PS3 Q3).

* A couple more notes on Circular motion:

• Recall we introduced angular velocity

$$\omega(t) = \frac{d\varphi}{dt}$$



• If $\omega(t)$ is constant then: $\omega = \frac{d\varphi}{dt}$

and so $\varphi(t) = \omega t$

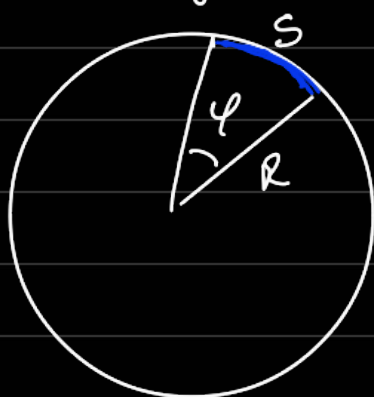
- We can introduce the period of the motion T

T : time it takes for the object to go around the circle once.

ex: The period of the earth is: $T_{\text{earth}} = 24 \text{ hours}$

→ It takes the earth ≈ 24 hours to make 1 full rotation

- Now we can relate the velocity V , the radius R and angular velocity



$$S = R\varphi$$

S : arc length

- Take the time derivative: $\frac{ds}{dt} = \frac{d}{dt}(R\varphi) = \frac{dR}{dt}\varphi + R\frac{d\varphi}{dt}$

→ Here R is constant so $\frac{dR}{dt} = 0$, so

$$\frac{ds}{dt} = R \frac{d\varphi}{dt} = R\omega$$

$$\omega \leftarrow \text{Recall: } \omega = \frac{d\varphi}{dt}$$

but $\frac{ds}{dt}$ is the velocity $\frac{ds}{dt} \leftarrow \begin{matrix} \text{change in distance} \\ \text{change in time} \end{matrix}$

$$\text{so } \frac{ds}{dt} = V:$$

$$V = R\omega$$

$$\text{thus, } \omega = \frac{V}{R}$$

- Now, if we go around the circle once the total distance traveled is equal to the circumference

of the circle

$$(\text{Circum.}) = 2\pi R$$

• If the period is T , then

$$\text{Velocity} \rightarrow V = \frac{2\pi R}{T} \quad \begin{array}{l} \leftarrow \text{Distance} \\ \leftarrow \text{time} \end{array}$$

• So, we have two different expressions for the vel.
→ Equating these two:

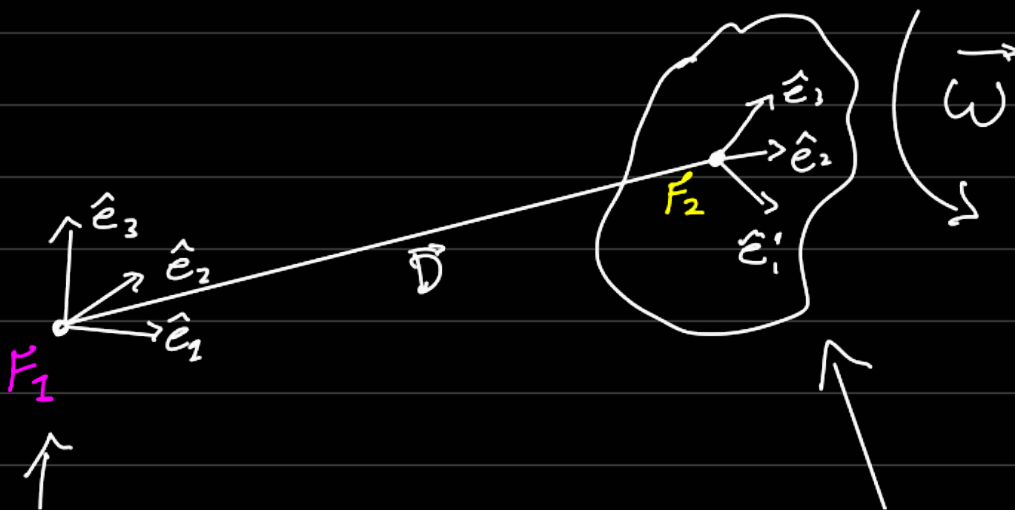
$$V = R\omega = \frac{2\pi R}{T} \Rightarrow \boxed{\omega T = 2\pi}$$

ω : Angular velocity
 V : "linear" velocity
 R : Radius
 T : Period

→ We'll need all of these concepts to solve Q3:

Set-up for Q3:

* For this problem we need to look at relative motion in terms of basis vectors and different reference frames:



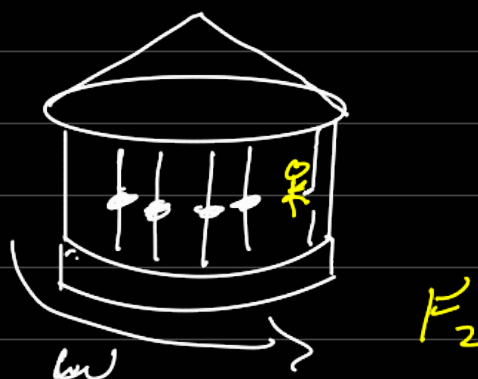
Reference frame 1:

- Uses basis $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$

Reference frame 2:

- Uses basis $(\hat{e}'_1, \hat{e}'_2, \hat{e}'_3)$
- Is rotating

ex: as an example we can think of ref 1 $[K_1]$ as a person standing on the ground and ref. 2 $[K_2]$ as a merry-go-round:



• merry-go-round rotates at angular velocity ω
 -> With this picture in mind let's delve into the math:

- Recall that in our previous session we found the rotating basis vectors: $(\hat{e}_r, \hat{e}_\phi)$ for polar coordinates

$$\hat{e}_r = \cos(\phi) \hat{e}_x + \sin(\phi) \hat{e}_y$$

$$\hat{e}_\phi = -\sin(\phi) \hat{e}_x + \cos(\phi) \hat{e}_y$$

what we found unique about this is that these basis vectors changed over time:

$$\frac{d\hat{e}_r}{dt} = \frac{d\phi}{dt} \frac{d\hat{e}_r}{d\phi}$$

chain-rule

- Our "usual" basis vectors $(\hat{e}_x, \hat{e}_y, \hat{e}_z)$ however, do not

$$\frac{d\hat{e}_x}{dt} = \frac{d\hat{e}_y}{dt} = \frac{d\hat{e}_z}{dt} = 0$$

• We call reference frame 1 an inertial reference frame

this is a frame which is not accelerating.

* Ref. frame 2 is a non-inertial frame: it is a frame which is accelerating.

→ Your notes call (reference frame 1) = S
(reference frame 2) = Σ

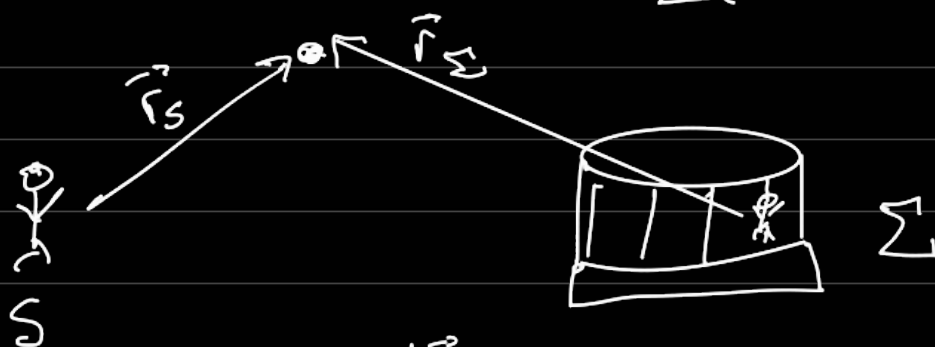
so we'll use this notation from now on (pg 35 of the lecture notes)

$$\vec{r}_S = x(t)\hat{e}_x + y(t)\hat{e}_y + z(t)\hat{e}_z$$

→ This is the position vector of some object in the inertial frame S (person standing)

$$\vec{r}_\Sigma = \xi(t)\hat{e}_\xi(t) + \eta(t)\hat{e}_\eta(t) + \zeta(t)\hat{e}_\zeta(t)$$

→ This is the position vector of some object in the non-inertial frame Σ (on the merry-go-round)



• The velocity $\vec{V}_S = \frac{d\vec{r}_S}{dt} = \dot{x}\hat{e}_x + \dot{y}\hat{e}_y + \dot{z}\hat{e}_z$ is simple in S

• However in Σ :

$$\begin{aligned}\vec{V}_\Sigma &= \frac{d\vec{r}_\Sigma}{dt} = \frac{d}{dt} \left(\xi(t)\hat{e}_\xi(t) + \dots \right) \\ &= \underbrace{\frac{d\xi}{dt}}_{\text{product rule}} \hat{e}_\xi(t) + \xi(t) \underbrace{\frac{d\hat{e}_\xi}{dt}}_{\text{Here the basis vectors change}} + \dots\end{aligned}$$

Here the basis vectors change

* this becomes:

$$\vec{V}_\Sigma = \left[\left(\dot{\xi}(t) \hat{e}_\xi(t) + \dot{\eta}(t) \hat{e}_\eta(t) + \dot{\zeta}(t) \hat{e}_\zeta(t) \right) + \left(\xi(t) \frac{d\hat{e}_\xi}{dt} + \eta(t) \frac{d\hat{e}_\eta}{dt} + \zeta(t) \frac{d\hat{e}_\zeta}{dt} \right) \right]$$

→ In this rotating frame we get an entire new term ()

• We find $\frac{d\hat{e}(t)}{dt}$ is related to the angular velocity:

$$\frac{d\hat{e}(t)}{dt} = \vec{\omega} \times \hat{e}(t)$$

• Plugging this back into ()

$$\begin{aligned} & \xi(t) (\vec{\omega} \times \hat{e}_\xi) + \eta(t) (\vec{\omega} \times \hat{e}_\eta) + \zeta(t) (\vec{\omega} \times \hat{e}_\zeta) \\ &= \vec{\omega} \times (\xi(t) \hat{e}_\xi) + \vec{\omega} \times (\eta(t) \hat{e}_\eta) + \vec{\omega} \times (\zeta(t) \hat{e}_\zeta) \\ &= \vec{\omega} \times (\xi(t) \hat{e}_\xi + \eta(t) \hat{e}_\eta + \zeta(t) \hat{e}_\zeta) \\ &= \vec{\omega} \times \vec{r}_\Sigma \end{aligned}$$

so, we found

$$\vec{V}_\Sigma = \vec{V}_{\text{apparent}, \Sigma} + \vec{\omega} \times \vec{r}_\Sigma$$

where $\vec{V}_{\text{apparent}, \Sigma} = \dot{\xi} \hat{e}_\xi + \dot{\eta} \hat{e}_\eta + \dot{\zeta} \hat{e}_\zeta$

which is the velocity we would have expected if this system was not rotating.

• If we take another derivative to find the acceleration, this is a bit involved so we'll just state the result: (see pages 40-44 of your notes)

$$a_\Sigma = a_{\Sigma, \text{apparent}} + \vec{\omega} \times \vec{r}_\Sigma(t) + \vec{\omega} \times (\vec{\omega} \times \vec{r}_\Sigma) + 2\vec{\omega} \times \vec{v}_{\Sigma, \text{app}}$$

